

Comparing Machine Learning Algorithms and Climate Data for Predicting Groundwater Levels: A Case Study of Kagera Basin in Rwanda

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Abstract

Predicting changes in groundwater levels is crucial for effective water management, particularly in regions facing climate change challenges. This study focuses on developing linear and non-linear predictive models for groundwater levels in wells in the Eastern Province of Rwanda. We evaluate four models: Multivariate Linear Regression (MLR), Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). These models leverage historical groundwater level data and weather data from two different sources: ERA5 Land and Visual Crossing.

Our analysis indicated that the forecasting results improve when using explanatory variables from Visual Crossing compared to those from ERA-5. Among the models evaluated, XGBoost demonstrates superior performance in predicting changes in GWLs. The XGBoost model, utilizing Visual Crossing weather data, achieved the highest accuracy in GWL prediction with an RMSE of **xx** and a coefficient of determination (R^2) of **xx**. Spearman and Pearson correlation analyses identified **xx** and **yy** variables as most influential on GWL for all wells analysed.

Furthermore, the project forecasts future groundwater level changes for 2025, revealing a projected decline of **x percent** compared to initial measurements. Such insights are crucial for guiding groundwater management strategies amidst anthropogenic climate change impacts. This research not only identifies the most suitable GWL modeling approach for data-limited

regions but also provides actionable forecasts to inform policy decisions regarding groundwater resource sustainability.

Keywords: Groundwater level, machine learning, Forecasting, ensemble models, Sub-Saharan Africa.

1. Introduction

Groundwater is a critical source supplying one-third of the world’s freshwater needs. The remaining 70% of freshwater comes from surface water sources like lakes, rivers, ice sheets, and glaciers [7]. In Sub-Saharan Africa (SSA), the dependence on groundwater is even more pronounced, with 75% of the population, as well as the industrial and agricultural sectors, relying on it [53, 16].

Given such high dependence, any risks to groundwater availability can significantly impact livelihoods. Additionally, due to hydroclimate variability driven by anthropogenic climate change and rapid population growth, groundwater vulnerability is becoming increasingly critical in SSA [21, 15].

Understanding groundwater levels in the Kagera River Basin is particularly crucial. The basin is of utmost ecosystem importance for East Africa as the groundwater originating here ultimately flows into the Nile River via Lake Victoria. The Nile River, in turn, serves as the lifeblood for 11 countries in Africa. Ecological concerns in this basin include deforestation, land use cover change, agricultural intensification, urbanization, wetland degradation, growing population and invasive species [50, 8].

Predicting groundwater levels in the Kagera Basin is essential to mitigate risks to water availability and sustain the ecological balance critical for the basin’s and Nile’s ecosystem. Despite its socio-ecological significance, there is a lack of long-term historical data on groundwater levels in the Kagera River Basin [46]. Coordinating such datasets is challenging due to the need for collaboration among institutions across four African countries. This absence of hydrological data for the whole Kagera River Basin hampers reliable groundwater level forecasts.

Nevertheless, some historical piezometric data is accessible for a sub-region of the Kagera River Basin in the Eastern Province of Rwanda. We augmented this dataset with climate reanalysis data. This study seeks to forecast groundwater dynamics in the province using exogenous variables,

thereby aiding the formulation of effective water management strategies for the region.

1.1. Literature Review

Forecasting groundwater levels is a critical challenge in hydrological studies. Groundwater levels are influenced by a dynamic interaction between recharge and discharge (outflow of groundwater to streams, lakes, oceans, or the atmosphere). This process is controlled by a variety of factors, including climate, soil properties, geological conditions, land cover, human water usage, and the surrounding ecosystems [43]. The complexity of these interactions, along with the scarcity of direct, long-term observations of groundwater levels and the limited understanding of local human-nature relationships, makes it difficult to predict how recharge rates vary over time and space and their relationship to climate.

Several numerical based models have been used for groundwater level prediction [44, 58, 17]. Open-source and proprietary software for such numerical based models include Finite Element subsurface Flow (FEFLOW) [1], Modular Finite Difference Groundwater Flow (MODFLOW) [52], and Saturated-Unsaturated Transport (SUTRA) [55]. These numerical models necessitate a vast array of parameters, such as aquifer properties, geology of the porous media, topography, boundary conditions, hydraulic conductivity, transitivity, recharge rates, chemical analysis, climate, and land use data [4]. There are two key advantages of numerical-based modeling. First, it provides a deep understanding of the underlying physical processes behind GWL fluctuations, thanks to well-defined mathematical methods. Some of the numerical models used are finite difference [37], finite volume [30], finite element [51], and element-free [40] methods. Second, it creates detailed representations of hydrological systems by integrating all available spatial and non-spatial data into a unified framework.

However, numerical-based models have inherent limitations. First, GWL prediction is computationally intensive and demanding of large hydrogeological survey data. Second, although traditionally these models can provide satisfactory GWL predictions [18, 9], their effectiveness can be limited by the complex, non-linear, and non-stationary characteristics of GWL fluctuations. Third, a practical limitation is that due to their resource-intensive nature and the historically unequal distribution of research funding between the Global North and South, there are few numerical-based GWL case studies in Sub-Saharan Africa [3, 13].

In regions where detailed hydrogeological survey data is lacking, combining classical statistics or machine learning forecasting with climate reanalysis data and historical GWL levels can facilitate initial case study analyses and GWL predictions.

Groundwater level prediction using classical statistical models [5] and machine learning (ML) models [10] has been successfully applied in various case studies. ML models, in particular, demonstrate robustness in handling complex datasets and effectively predicting non-linear systems. For a comprehensive review, see [45]. However, there remains a scarcity of studies utilizing machine learning for GWL prediction in Sub-Saharan Africa, with notable exceptions such as studies in Nigeria [2] and Kombo [32].

Researchers have explored diverse ML models for predicting GWL, addressing various challenges, using different datasets and methodologies. Traditional ML models such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Adaptive Neuro-Fuzzy Inference Systems (ANFIS) have been pivotal when there is no availability of the entire physical description [34].

Zhang et al. [57] conducted a study or experiment comparing LSTM (Long Short-Term Memory) models with traditional Feed-Forward Neural Networks (FFNN). The study found that LSTM models performed better than FFNN when predicting GWL within areas used for agricultural purposes. As explained in [45] most ML groundwater level predictions predominantly utilize input parameters derived from historical groundwater level data, focusing on autocorrelated variables. However, they have limited integration of multi-time series exogenous meteorological and hydrological data [6].

This problem may be addressed by seasonal autoregressive integrated moving averages model with exogenous data (SARIMAX) models, which incorporate exogenous leading variables to account for additional information affecting variables not contained in the GWL historical time series data; examples of such variables include evapotranspiration, sea level pressure and humidity. SARIMAX has been widely employed in hydrology forecasting [23].

In a comparative study, [28] assessed the predictive performance of multilinear regression, random forest, and extreme gradient boosting for forecasting groundwater levels in Indonesian tropical peatlands. Their findings revealed that extreme gradient boosting surpassed the other models in accuracy. Drawing inspiration from this research, we extend this comparative

analysis and incorporate SARIMAX as an additional statistical model to be evaluated in the context of groundwater level prediction in the Eastern Province of Rwanda.

This article addresses the following research question:

- How do multivariate classical linear statistical methods and non-linear machine learning algorithms compare in accurately and reliably forecasting point groundwater levels using piezometric data from different time horizons in the Eastern Province of Rwanda?
- What critical factors affect how well linear statistical methods and non-linear machine learning algorithms predict groundwater levels?
- How do prediction outcomes differ when using weather data from Visual Crossing compared to ERA5 Land for forecasting groundwater levels?

To answer this question, we gathered historical groundwater level data from four wells in Rwanda’s Eastern Province, supplemented by two climate datasets. We conducted a thorough analysis of data completeness and performed interpolation of missing data points to ensure the robustness of our datasets. Subsequently, we developed and trained four distinct models for each well’s dataset to capture linear and nonlinear relationships among multiple time series. We evaluated the accuracy of these models for point predictions of groundwater levels and analyzed their performance across different periods of available data.

This study advances previous research by incorporating a broader range of meteorological variables that influence groundwater levels, including factors such as evapotranspiration (ET), humidity, and sea level. Additionally, it offers a detailed comparison between linear and nonlinear forecasting models, providing new insights into their relative accuracy. By examining the dependencies and interactions among these variables, analyzing different data horizons, and addressing various data completeness scenarios, the study enhances our understanding of how these factors collectively impact groundwater levels. Lastly, to the best of our knowledge, no previous study has compared the ERA-5 Land with the Visual Crossing weather dataset in this context, making this comparison particularly valuable.

2. Materials and Methods

2.1. Study Region

This study concentrates on the East Province of Rwanda. The Eastern Province's hydrogeographical importance is highlighted by its inclusion in the Kagera River Basin, a key hydrological system in East-Central Africa (see Figure 1).

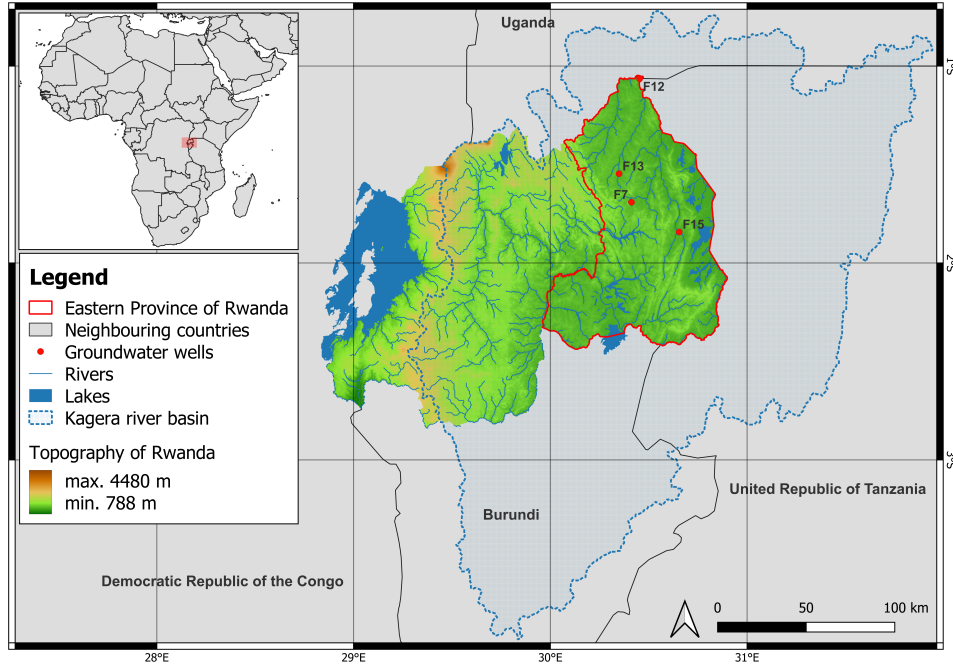


Figure 1: Location of groundwater wells inside the Kagera Basin in the East Province of Rwanda.

The province hosts vital water bodies such as the Bugesera Lakes and those within Akagera National Park, which is a biodiversity hotspot under climatic threat [31, 49]. These contribute significantly to the hydrology of the Kagera River Basin. The Kagera River, originating from the highlands of East-Central Africa, is approximately 400 km long. Upon reaching the Rwanda-Tanzania border, it flows north along this boundary, then turns eastward, crossing the lowland floodplains of Tanzania and Uganda, before finally emptying into Lake Victoria at an elevation of 1,145 meters. The Kagera basin covers about 59,800 km^2 across Rwanda, Tanzania, Uganda,

and Burundi, supporting millions whose livelihoods depend on agriculture and fishing [25].

Annual rainfall averages 600–800 mm/year, with notable variation across the region [42]. The Eastern province faces recurring droughts, impacting agriculture and water access, particularly during dry periods [56]. In 2016, drought affected the Eastern Province, leading to food insecurity among 225,000 people [56].

The map in Figure 1 was created with QGIS [38] using various datasets from multiple sources. The subnational administrative boundaries of Rwanda and water bodies were sourced from the Humanitarian Data Exchange (HDX) provided by [47] and [48] respectively. The shapefile for the Eastern province of Rwanda and the Kagera basin were taken from [39] and [24] respectively. Lastly, the topographic information with a 90-meter resolution for Rwanda was accessed via [41].

2.2. Datasets

2.2.1. Groundwater wells

Groundwater level (GWL) data was collected from the Rwanda Water Portal (RWP) [42] at daily intervals over varying periods. The Eastern Province exhibited distinct GWL patterns, including fluctuations, increases, and decreases. Four wells—F12, F13, F15, and F17—were selected for this study to represent the region’s hydrological variability, ranging from frequent short-term fluctuations to more stable trends. While most fluctuations remained within a narrow range, characteristic of GWL stability over time, the peaks and troughs warrant further analysis. As shown in Figure 2, wells F13 and F15 display more frequent fluctuations compared to F12 and F17, suggesting greater variability likely due to localized extraction or recharge patterns. The dynamic changes in GWL, particularly in F13 and F15, may be influenced by short-term factors such as rainfall variability, seasonal recharge, or groundwater extraction. In contrast, well F17, despite an initial drop, shows relatively stable GWL over time, indicating differing hydrological behaviors across the wells.

Table 1 provides details on the selected monitoring wells and their characteristics. The depths of the wells vary between 37 and 77 meters. Data collection periods range from 291 (F12) days to 1497 days (F15). The standard deviations are all quite low, suggesting that the GWLs are relatively stable. The mean Water Column Height (WCH) reflects the volume of accessible water. A taller water column generally signifies a greater quantity

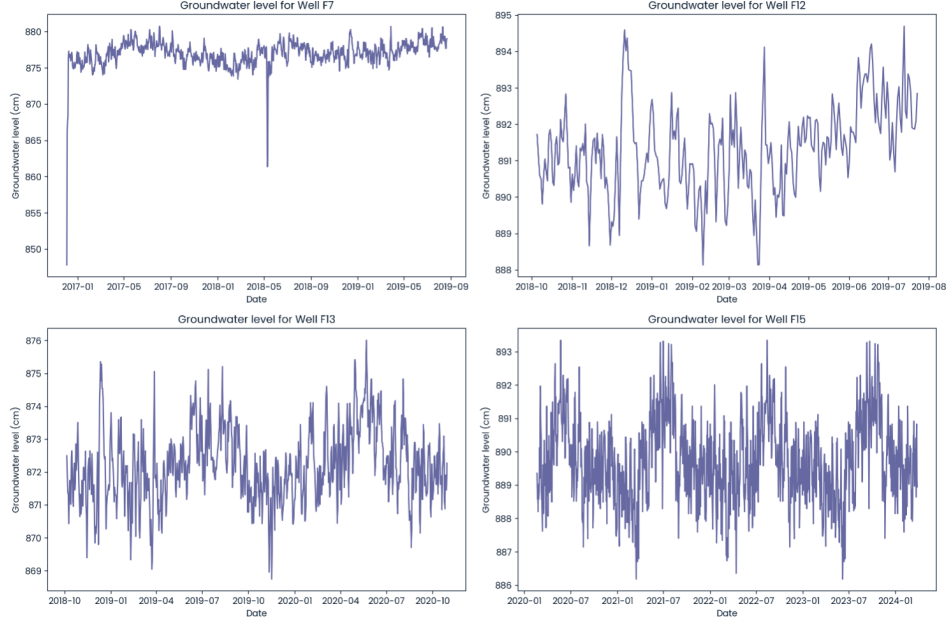


Figure 2: Groundwater Level Trends Across Selected Wells in East Rwanda

of accessible groundwater [22]. Here, WCH varies significantly, from 28.1 meters (F7) to 68.31 meters (F13).

Table 1: Target Variable Characteristics: Groundwater Levels per Well

ID	Name	Depth	Period	Mean GWL	Stand. Dev.	Mean WCH
<i>F7</i>	Gatisbo- Rugarama	37 m	991 days	877.87 cm	1.85	28.1 m
<i>F12</i>	Nyagatare- Matimba	60 m	291 days	891.32 cm	1.21	51.08 m
<i>F13</i>	Gatsibo- Kabarore	77 m	755 days	869.02 cm	1.13	68.31 m
<i>F15</i>	Kayonza- Mwiri	55 m	1497 days	889.74 cm	1.25	46.1 m

2.2.2. ERA-5 Dataset

In this study, we utilized the fifth generation of the European ReAnalysis (ERA5) land component, referred to as ERA5-Land [35]. The ERA5 dataset provides hourly estimates of climate variables by integrating data from a

variety of sources, including satellite measurements of ocean wind speed, land humidity, and observations from weather balloons, aircraft, and ground stations [27]. While ground-based measurements are valuable, their global distribution is uneven. To address gaps where direct observations are limited or unavailable, ERA5 employs advanced numerical weather prediction models [11].

We retrieved ERA5 data using the GPS coordinates of each groundwater well through Climate Engine [20]. The dataset provided hourly information on land surface variables such as mean, maximum, and minimum temperatures, along with atmospheric variables such as precipitation, wind speed, and net solar radiation. Additionally, evapotranspiration was included, calculated using the Hargreaves model [26]. The spatial resolution for all variables is approximately 11.1 km ($0.125^\circ \times 0.125^\circ$). The downloaded data corresponded to four distinct study periods, each for one of the four wells. To align with the temporal resolution of groundwater data from RWP, we aggregated the hourly ERA5 data into daily summaries.

2.3. Visual Crossing Dataset

Another dataset used for the explanatory variables of groundwater was the Visual Crossing (VC) Weather Data [54]. This dataset provides historical weather information for locations worldwide by aggregating data from over 100,000 global weather stations, satellites, and other sources. The data is processed through an algorithm that considers the three closest weather stations within a 50 km radius of the target location. Records from these stations are combined using a weighted mean, giving more influence to closer stations (within 10 km). The dataset is normalized for reporting times and adjusted for time zones to ensure consistency. Users can customize the data by adjusting parameters such as the number of stations included, the search radius, and the time resolution of the variables. Our dataset from VC provided daily information on: precipitation (mm), mean temperature ($^\circ\text{C}$), max temperature ($^\circ\text{C}$), and min temperature ($^\circ\text{C}$), and wind speed, and net solar radiation.

Figure 3 portrays the importance of contrasting the two datasets, which while containing several similar variables, exhibit differences. The VC dataset shows more frequent dry spells in precipitation and a narrower temperature range between 20°C and 22°C . In contrast, the ERA5 dataset indicates more consistent but lighter rainfall and a broader temperature distribution ranging from 19°C to 24°C . The windspeed data from VC peaks around 3.5–4

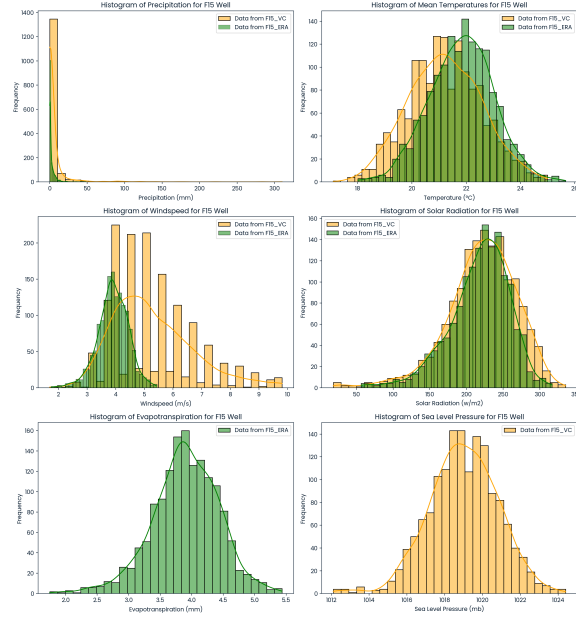


Figure 3: Comparison of Independent Variables for GWL in Well F15 Using Visual Crossing and ERA5 Land Data.

m/s, while ERA5 records a wider distribution, with speeds reaching 9 m/s. Similarly, solar radiation data from ERA5 exhibits greater variability with a peak near 250 W/m^2 , compared to VC's skewed distribution around 150 W/m^2 . Evapotranspiration, present only in ERA5, is concentrated between 3–4.5 mm/day, whereas sea-level pressure, available solely in VC, is narrowly distributed between 1018 and 1022 mb. These differences highlight the complementary nature of each dataset and their potential to yield varying outcomes in groundwater level prediction.

3. Study Focus

This study aims to compare the performance of multivariate classical linear statistical methods and non-linear machine learning algorithms in forecasting groundwater levels using piezometric data from the Eastern Province of Rwanda. The focus is on evaluating the accuracy and reliability of these methods across varying time horizons.

The study has three main focus. First, **model comparison**, where both linear methods (Multivariate Linear Regression and SARIMAX) and non-

linear ensemble learning models (Random Forest and XGBoost) are applied to forecast groundwater levels. Accuracy will be assessed using metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). Second, **time horizon sensitivity**, involving the analysis of groundwater levels from four wells with varying time series lengths (e.g., 291 days for short-term and 1497 days for long-term intervals). This analysis will assess how the predictive performance of each method changes with respect to different forecasting horizons.

Third, the study will **compare weather datasets** for groundwater level forecasting, using data from Visual Crossing and ERA5 Land, which differ in resolution, spatial coverage, and the set of available variables. To determine which dataset yields better model performance, we will apply feature importance techniques (**SHAP values**) and statistical tests to identify the most significant variables. Once these significant variables are identified from each dataset, we will combine them to create enriched feature sets that include statistically significant or unique variables from both sources. Models will then be retrained on these combined datasets, and error metrics will be compared to evaluate potential improvements in predictive performance.

This research aims to determine the optimal methods and weather data sources for reliable groundwater forecasting.

3.1. Multi-linear Regression

We first use Multi-linear regression (MLR) to determine the statistical relationships between the dependent and multiple independent or predictor variables [36]. MLR are fitted by the least squares approach [33]. The general form of MLR is as follows:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_i x_i + \epsilon$$

where y is the dependent variable and x_i are the i distinct independent or predictor variables. β_0 is the intercept term. All the other regression coefficients β_i represent the expected change in the predicted variable y per unit change in x_i when all the remaining regression coefficients are held constant. ϵ is the error term, representing the variability in y that is not explained by the predictors.

3.2. SARIMAX

The SARIMAX method is an extension of the classic Autoregressive Integrated Moving Average (ARIMA) method for working with seasonal

time series and taking into account multiple exogenous variables. A typical ARIMA(p,d,q) model can be expressed by:

$$\varphi_p(B)(1-B)^d z_t = \theta_q(B)\varepsilon_t \quad (1)$$

where z_t represents a nonstationary time series at time t , ε_t is a white noise, d is the order of differencing, B is the backward shift operator defined by $Bz_t = z_{t-1}$ and $\varphi_p(B)$ is the autoregressive operator defined as:

$$\varphi_p(B) = 1 - \varphi_1 B - \varphi_2 B^2 - \dots - \varphi_p B^p. \quad (2)$$

Also, $\theta_q(B)$ is the moving average operator defined as:

$$\theta_q(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q \quad (3)$$

The seasonal ARIMA model (SARIMA) $(p, d, q), (P, D, Q)$, where p, d, q and P, D, Q are non-negative integers that refer to the polynomial order of the autoregressive (AR), integrated (I), and moving average (MA) parts of the non-seasonal and seasonal components of the model. The SARIMA model is described mathematically as:

$$\varphi_p(B)\phi_p\nabla^d\nabla_s^D y_t = \theta_q(B)\Theta_Q(B^s)\epsilon_t$$

where y_t is the predicted variable (GWL at time t), $\varphi_p(B)$ is the regular AR polynomial of order p , θ_q is the regular MA polynomial of order q , $\phi_P(B^s)$ is the seasonal AR polynomial of order P and $\Theta_Q(B^s)$ is the seasonal MA polynomial of order Q . The differentiating operator ∇^d and the seasonal differentiating operator ∇_s^D eliminate the non-seasonal and seasonal non-stationarity. B is the backshift operator (i.e. $B^k(y_t) = y_{t-k}$). The term ϵ_t follows a white noise process and s defines the seasonal period. The model development was based on the Box-Jenkins methodology [12], which consists of four iterative steps: 1) identification, 2) estimation, 3) diagnostic checking and 4) forecasting.

The multivariate version of the SARIMA model, called Seasonal ARIMA with eXogenous factors (SARIMAX) is generally expressed as:

$$\varphi_p(B)\phi_p\nabla^d\nabla_s^D y_t = \beta_k x_{k,t} + \theta_q(B)\Theta_Q(B^s)\epsilon_t$$

where $x_{k,t}$ is the vector including the k th explanatory input variables at time t and β_k is the coefficient value of the k -th exogenous variable. The

stationarity and invertibility conditions are equal to those of the ARMA models.

In our SARIMAX model, the parameters were determined by analyzing the autocorrelation function (ACF) and partial autocorrelation function (PACF), which provided upper bounds for the p and q values of the time series. Additionally, a grid search was performed to minimize the Akaike Information Criterion (AIC).

3.3. SARIMAX Bayesiano

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3.4. Random Forest

Random forest (RF) is an ensemble learning method composed of decision trees. During training, a large number of decision trees are built, where each tree is trained on a random subset of the training data (bootstrap samples) and uses a random subset of features (feature bagging). The final prediction of the random forest is typically the average prediction of all individual trees for regression tasks or a majority vote for classification tasks.

Groundwater level (GWL) prediction models involve multiple predictor variables, such as climate data and geographical features. Random forests can effectively handle these large feature sets without requiring prior feature selection or dimensionality reduction, as they automatically select relevant features during tree construction.

In RF regression models for GWL prediction, each tree in the forest is a regression tree where non-leaf nodes contain decision rules and leaf nodes represent predictions. The parameters defining RF include the number of trees in the forest B and the number of features used to split nodes D .

RF works as [14]:

- From the initial data set X , multiple bootstrap samples X_i (where i is the bootstrap iteration) are randomly selected. Each bootstrap sample consists of two-thirds of the data, and the remaining one-third serves as out-of-bag samples. Each tree in the forest is trained using a specific bootstrap sample.
- The input variable that minimizes the mean squared error is chosen as the splitting criterion for each node in the regression tree. A new split is selected randomly from all available features.

- This process of recursively splitting the data at each internal node continues until the trees reach a predefined stopping condition.
- Given B as the total number of trees in the forest and T_b as each individual tree, the final regression predictions are computed as:

$$\hat{y}(x_i) = \frac{1}{n} B \sum_{b=1}^B T_b(x_i) \quad (4)$$

This formula averages the predictions from all trees T_b to produce the final prediction $\hat{y}(x_i)$ for each input x_i .

3.5. XGBoost

XGBoost is a powerful gradient boosting algorithm used for classification, regression, and ranking tasks by combining predictions from multiple weak models, typically decision trees, in an additive manner [29]. It is favored for its speed, efficiency in handling large datasets, and superior predictive quality.

When applying XGBoost to predict groundwater levels (GWLs), feature engineering plays a critical role in enhancing predictive capability. Careful evaluation of the dataset helps identify features that significantly contribute to the prediction task. Adjusting existing features or creating new ones can optimize model performance. Additional features derived from climate data, such as hour, day of the week, month, year, and week of the year, can further improve accuracy.

The workflow of XGBoost is depicted in Figure ???. For a dataset D with n examples and m features representing climate variables and additional features XGBoost operates as [19].

The predicted output $\hat{y}_i$ from an ensemble of K trees in the model is given by:

$$\hat{y}_i = \phi(x) = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (5)$$

To optimize predictions, XGBoost minimizes an objective function combining loss l and regularization Σ :

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad (6)$$

where $l = \hat{y}_i - y_i$ is the loss function. $\Omega(f_k)$ is the complexity of the model, which aids in preventing overfitting. Here, T stands for the number of leaves in the tree, and w denotes the weight assigned to each leaf.

Boosting iteratively improves the model by adding new functions (trees) in each iteration t :

$$L^{(t)} = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_k) \quad (7)$$

Using a second-order approximation, XGBoost optimizes the objective function efficiently. The performance of XGBoost and other models is evaluated using various metrics, ensuring robust assessment and comparison.

4. Evaluation

In this paper, the prediction model is evaluated using three error evaluation metrics. In the following equation, N denotes the number of observations, y_i denotes the actual observed groundwater level on day i , $\hat{y}_i$ denotes the predicted groundwater level on day i , where $\bar{y}$ is the mean of the actual observed groundwater levels, and T is the length of the historical data used for training the model .

Mean Absolute Scaled Error (MASE) compares the error of a forecasting model to the error of a naive model, typically one that uses the last observed value as the prediction. It indicates whether the model performs better or worse than the benchmark. A value of 1 denotes equivalent performance to the naive model; values less than 1 suggest improvement, while values greater than 1 indicate inferiority. MASE is defined as:

$$\text{MASE} = \frac{\frac{1}{N} \sum_{i=T+1}^{T+N} |y_i - \hat{y}_i|}{\frac{1}{T-1} \sum_{i=2}^T |y_i - y_{i-1}|}. \quad (8)$$

Root Mean Square Error (RMSE) is used to evaluate the deviation of the predicted value from the actual value, and the range is $[0, +\infty)$, with a smaller value indicating a higher prediction accuracy.

$$\text{RMSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (9)$$

Coefficient of Determination (R^2) explains how well the predictions from a model fit the actual data. It represents the proportion of variance in the

dependent variable that is predictable from the independent variables. An R^2 value of 1 indicates perfect prediction, while a value of 0 indicates that the model does no better than the mean of the actual values. Negative values of R^2 can occur when the model fits the data worse than a horizontal line (the mean of the data). The formula for R^2 is:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}. \quad (10)$$

Mean Absolute Percentage Error (MAPE) measures the accuracy of a prediction model by calculating the average absolute percentage difference between predicted and actual values. MAPE is unit-free and is expressed as a percentage, making it easy to interpret and compare across datasets. However, it is sensitive to small values, as division by small actual values can lead to large percentage errors. It is defined as:

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100. \quad (11)$$

5. Results and Discussion

The results of the forecasts using different datasets and four forecasting methods are presented in Table 3.

This study employs linear models (MLR and SARIMAX) and ensemble machine learning methods (RF and XGBoost) to predict groundwater levels (GWL). The statistical metrics MASE, RMSE, and R^2 are used to evaluate the prediction models, as presented in Tables 3 and 4. Upon reviewing the results in Tables 3 and 4, it is evident that the SARIMAX model outperforms the other models. Additionally, the Visual Crossing (VC) dataset proves to be more effective than ERA5 Land for predicting GWLs. However, the best overall predictions are achieved by combining the VC dataset with sea level data from ERA5 Land.

RE-write RED IN THE END. Overall, the VC dataset performs better than the ERA5-Land dataset across all wells, during the test period. The VC model shows fewer deviations from the actual GWL and captures the overall trend more accurately. Among the wells, Well F12 and Well F15 show the best performance with the VC dataset, with the predictions closely following the actual GWL.

Table 2: Summary of Key Differences between Performance Metrics for GWL Prediction

Metric	Sensitivity to Outliers	Scale-Dependent	Unit	Intuitiveness	Application Suitability for GWL Prediction (in cm)
MASE	Low	No	Dimensionless	Medium	Ideal for GWL prediction due to low variability and ability to compare to a naive model
RMSE	High	Yes	cm	Medium	Good for highlighting large deviations in GWL, measured directly in cm, making it easy to interpret errors
R²	N/A (measures variance)	No	Dimensionless	High (explains variance)	Useful for understanding how well the model explains GWL variations, especially in low-variance data
MAPE	Low (but sensitive to small values)	No	Percentage	High	Caution needed due to low GWL variability; provides a simple percentage-based error measure, but may overestimate small errors. Therefore, this usual metric is not applied in this study.

5.0.1. MLR results

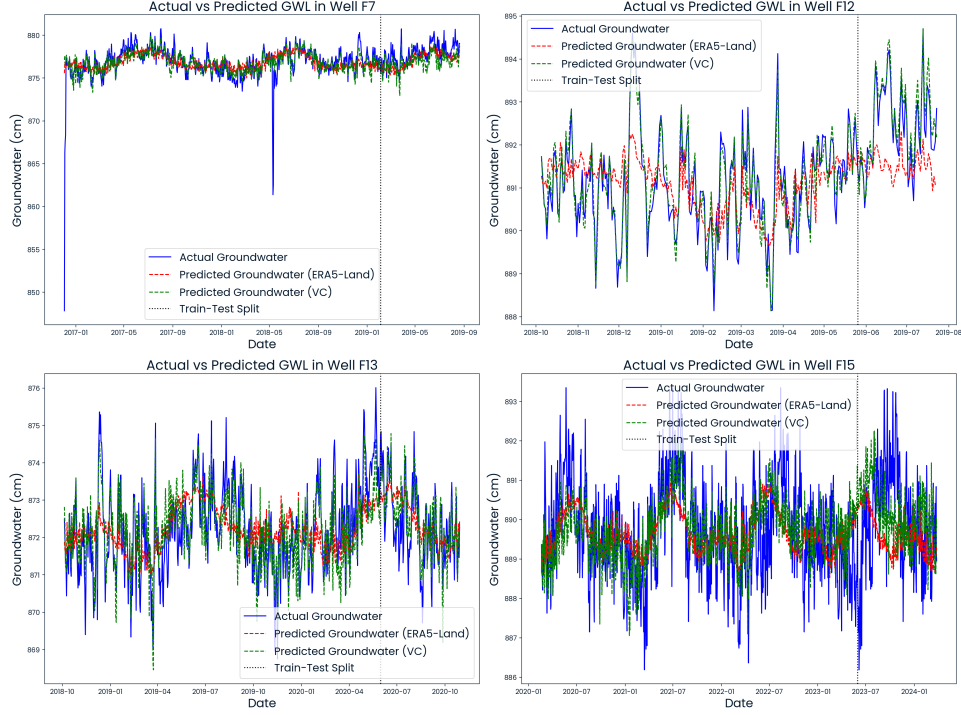


Figure 4: Comparison of Actual and Predicted GWLs across Four Wells using MLR with Visual Crossing (VC) and ERA5 Land Data Weather Data Inputs.

The Figure 4 presents the actual and predicted GWL for four wells over their different time periods. The blue line represents the measured actual groundwater levels. Predicted GWL values are derived using two weather data sources: ERA5-Land (red dashed line) and Visual Crossing (green dashed line), both of which serve as external linear regressors in the prediction models. The vertical dotted line marks the train-test split, with 80% of the data used for training and 20% for testing the model performance.

The MLR models' performance is influenced by various weather-related variables, with some regressors being more significant than others in predicting GWL. The significance is determined by p-values, with lower p-values indicating higher significance. Across all wells, precipitation, evapotranspiration, solar radiation and sea-level are the most consistently significant external regressors across the wells. These variables play a key role in influencing overall groundwater dynamics.

Analyzing Table 3 and Figure 4 MLR best performance is observed for Well F12 using VC, where the MLR model achieves an R^2 of 0.847, low RMSE, and low MASE, indicating that the VC dataset is better suited to this well for MLR modeling. Well F13 also shows good performance with VC ($R^2 = 0.650$), though performance drops significantly with ERA5-Land. For well F7, the MLR model performs poorly with both datasets, but ERA5-Land shows a slightly higher R^2 . Well F15 has poor performance with both datasets, with ERA5-Land providing a marginally better R^2 than VC. VC generally performs better in Well F12 and Well F13, offering stronger predictive capability for groundwater levels. ERA5-Land shows slight advantages in Well F7 and Well F15, but overall, both datasets have low predictive accuracy in these wells. The MLR model's success is highly well-dependent, with different datasets excelling in different locations.

5.1. SARIMAX results

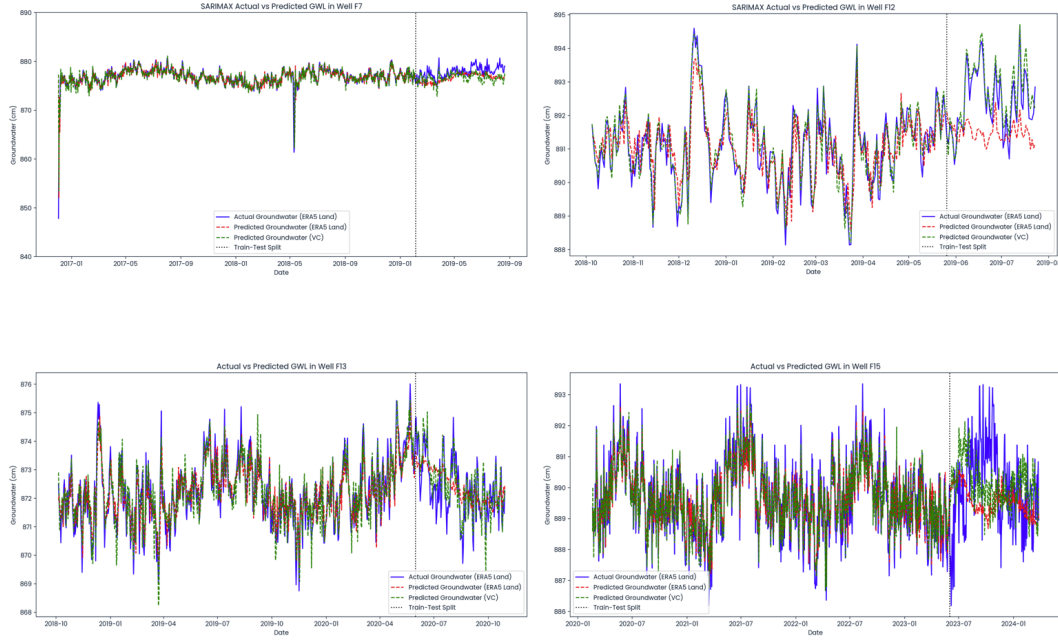


Figure 5: Comparison of Actual and Predicted GWLs across Four Wells using SARIMAX with VC and ERA5 Land Data Weather Data Inputs.

SARIMAX performs best in Well F12 with VC dataset and Well F13 (both datasets). In these cases, the R^2 values are reasonably high (0.86 in

F12 and 0.72 in F13 for Visual Crossing). SARIMAX performs poorly in Well F7 and F15, especially in the ERA5-Land dataset. The negative R^2 values (shown in Table 3) in these cases indicate that the model performs worse than a simple mean prediction.

5.1.1. *SARIMAX Bayesiano*

5.1.2. *Random Forest*

5.1.3. *XGBoost*

6. Conclusion

The hydrogeological system around well F12 could be simpler, meaning fewer external factors complicate the relationship between groundwater levels and weather. In some regions, the geology or hydrology is more straightforward, and groundwater responds more directly to rainfall, temperature, and evaporation, making such systems easier to model using linear regressions like MLR. In contrast, wells with more complex systems, such as interactions with rivers, multiple aquifers, or human interference (e.g., extraction or irrigation), can introduce non-linearities and confounders that make prediction with simpler models more difficult.

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Dataset	Well	Model	MASE	RMSE	R^2
Visual Crossing	F7	MLR	1.51	1.19	0.07
	F7	SARIMAX	1.93	1.44	- 0.58
	F7	RF			
	F7	XGBoost			
	F12	MLR	0.48	0.36	0.85
	F12	SARIMAX	0.49	0.36	0.86
	F12	RF			
	F12	XGBoost			
	F13	MLR	0.796	0.613	0.650
	F13	SARIMAX	0.73	0.55	0.72
	F13	RF			
	F13	XGBoost			
	F15	MLR	1.87	1.58	-0.450
	F15	SARIMAX	1.85	1.54	-0.38
	F15	RF			
	F15	XGBoost			
ERA5-Land	F7	MLR	1.51	1.19	0.155
	F7	SARIMAX	2.05	1.54	-0.81
	F7	RF			
	F7	XGBoost			
	F12	MLR	1.69	1.24	-0.80
	F12	SARIMAX	1.74	1.28	-0.915
	F12	RF			
	F12	XGBoost			
	F13	MLR	1.267	0.922	0.208
	F13	SARIMAX	1.22	0.88	0.28
	F13	RF			
	F13	XGBoost			
	F15	MLR	1.88	1.59	0.17
	F15	SARIMAX	1.88	1.57	-0.43
	F15	RF			
	F15	XGBoost			

Table 3: Model performance on VC and ERA5 Land datasets for wells F7, F12, F13, and F15

Dataset	Well	Model	MASE	RMSE	R^2
VC +ET	F7	MLR			
	F7	SARIMAX			
	F7	RF			
	F7	XGBoost			
	F12	MLR			
	F12	SARIMAX			
	F12	RF			
	F12	XGBoost			
	F13	MLR			
	F13	SARIMAX	0	0	0
	F13	RF			0
	F13	XGBoost			
	F15	MLR			
	F15	SARIMAX			
	F15	RF			
	F15	XGBoost			
ERA5-Land +SeaLevel	F7	MLR			
	F7	SARIMAX			
	F7	RF			
	F7	XGBoost			
	F12	MLR			
	F12	SARIMAX			
	F12	RF			
	F12	XGBoost			
	F13	MLR			
	F13	SARIMAX			
	F13	RF			
	F13	XGBoost			
	F15	MLR			
	F15	SARIMAX			
	F15	RF			
	F15	XGBoost			

Table 4: Model performance on combined datasets for wells F7, F12, F13, and F15